Writing in Computer Science: Math

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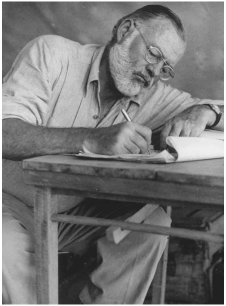
Outline:

1. Motivation
2. Writing in the language of Arithmetics
3. Adding English
4. Writing Scientific Documents

Why Technical Writing is Hard?: writing is hard

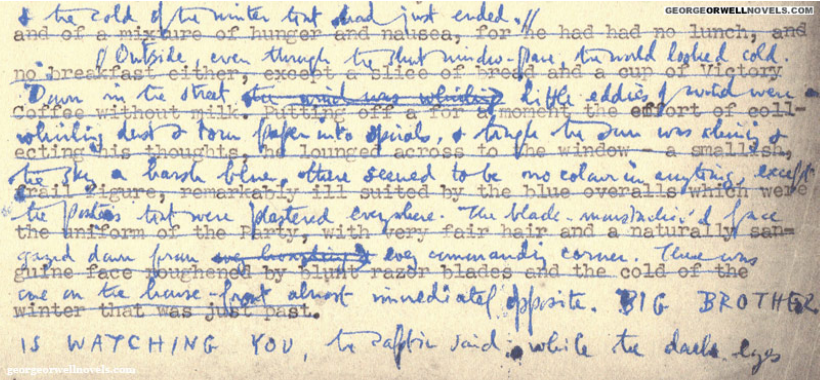
E. Hemingway

There is nothing to writing. All you do is sit down at a typewriter and bleed



Why Technical Writing is Hard?: Non-sequential

Writing is thinking. To write well is to think clearly.



& the cold of the winter that had just ended. //
~~and of a mixture of hunger and nausea, for he had had no lunch, and~~
// Outside, even through the shut window-pane, the world looked cold.
~~no breakfast either, except a slice of bread and a cup of Victory~~
Down in the street ~~the wind was whirling~~ little eddies of wind were
~~Coffee without milk. Putting off a for a moment the effort of coll-~~
whirling dust & torn paper into spirals, & though the sun was shining &
~~ecting his thoughts, he lounged across to the window - a smallish,~~
the sky a harsh blue, there seemed to be no colour in anything, except
~~frail figure, remarkably ill suited by the blue overalls which were~~
the posters that were plastered everywhere. The black-moustachio'd face
~~the uniform of the Party, with very fair hair and a naturally san-~~
gazed down from ~~every commanding~~ every commanding corner. There was
~~guine face roughened by blunt razor blades and the cold of the~~
one on the house-front almost immediately opposite. BIG BROTHER
~~winter that was just past.~~
IS WATCHING YOU, the caption said, while the dark eyes

Fragment from first page of 1984's draft by George Orwell

Why Technical Writing is Hard?: precision and simplicity

Stewart Stafford, poet

Never confuse precision with simplicity - the best people make highly-complex things seem easy through natural ability.

Why Technical Writing is Hard?: Little practice

During education, engineers listen to and read a lot of technical stuff, but they speak out and write little

Why Technical Writing is Hard?: The Curse of Knowledge

It is difficult to imagine what it's like to not know what you know.

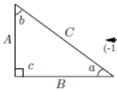
Curse of Knowledge

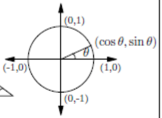
Cognitive bias where a person with specialized knowledge assumes that others, who have less of it, can easily understand the topic.

Writing Math

Theoretical Computer Science Cheat Sheet

Trigonometry





Pythagorean theorem:
$$C^2 = A^2 + B^2.$$

Definitions:
$$\sin a = A/C, \quad \cos a = B/C,$$
$$\csc a = C/A, \quad \sec a = C/B,$$
$$\tan a = \frac{\sin a}{\cos a} = \frac{A}{B}, \quad \cot a = \frac{\cos a}{\sin a} = \frac{B}{A}.$$

Area, radius of inscribed circle:
$$\tfrac{1}{2}AB, \quad \frac{AB}{A+B+C}.$$

Identities:
$$\sin x = \frac{1}{\csc x}, \qquad \cos x = \frac{1}{\sec x},$$
$$\tan x = \frac{1}{\cot x}, \qquad \sin^2 x + \cos^2 x = 1,$$
$$1 + \tan^2 x = \sec^2 x, \qquad 1 + \cot^2 x = \csc^2 x,$$
$$\sin x = \cos\left(\tfrac{\pi}{2} - x\right), \qquad \sin x = \sin(\pi - x),$$
$$\cos x = -\cos(\pi - x), \qquad \tan x = \cot\left(\tfrac{\pi}{2} - x\right),$$
$$\cot x = -\cot(\pi - x), \qquad \csc x = \cot\tfrac{x}{2} - \cot x,$$
$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y,$$
$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y,$$
$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y},$$

Matrices

Multiplication:
$$C = A \cdot B, \quad c_{i,j} = \sum_{k=1}^{n} a_{i,k} b_{k,j}.$$

Determinants: $\det A \neq 0$ iff A is non-singular.
$$\det A \cdot B = \det A \cdot \det B,$$
$$\det A = \sum_{\pi} \prod_{i=1}^{n} \operatorname{sign}(\pi) a_{i,\pi(i)}.$$

2×2 and 3×3 determinant:
$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc,$$
$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = g\begin{vmatrix} b & c \\ e & f \end{vmatrix} - h\begin{vmatrix} a & c \\ d & f \end{vmatrix} + i\begin{vmatrix} a & b \\ d & e \end{vmatrix}$$
$$= \begin{array}{c} aei + bfg + cdh \\ - ceg - fha - ibd. \end{array}$$

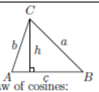
Permanents:
$$\operatorname{perm} A = \sum_{\pi} \prod_{i=1}^{n} a_{i,\pi(i)}.$$

Hyperbolic Functions

Definitions:
$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2},$$
$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \qquad \operatorname{csch} x = \frac{1}{\sinh x},$$
$$\operatorname{sech} x = \frac{1}{\cosh x}, \qquad \coth x = \frac{1}{\tanh x}.$$

Identities:
$$\cosh^2 x - \sinh^2 x = 1, \qquad \tanh^2 x + \operatorname{sech}^2 x = 1,$$
$$\coth^2 x - \operatorname{csch}^2 x = 1, \qquad \sinh(-x) = -\sinh x,$$
$$\cosh(-x) = \cosh x, \qquad \tanh(-x) = -\tanh x,$$
$$\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y,$$

More Trig.



Law of cosines:
$$c^2 = a^2 + b^2 - 2ab\cos C.$$

Area:
$$A = \tfrac{1}{2}hc,$$
$$= \tfrac{1}{2}ab\sin C,$$
$$= \frac{c^2 \sin A \sin B}{2\sin C}.$$

Heron's formula:
$$A = \sqrt{s \cdot s_a \cdot s_b \cdot s_c},$$
$$s = \tfrac{1}{2}(a+b+c),$$
$$s_a = s - a,$$
$$s_b = s - b,$$
$$s_c = s - c.$$

More identities:
$$\sin\tfrac{x}{2} = \sqrt{\frac{1-\cos x}{2}},$$
$$\cos\tfrac{x}{2} = \sqrt{\frac{1+\cos x}{2}},$$
$$\tan\tfrac{x}{2} = \sqrt{\frac{1-\cos x}{1+\cos x}},$$
$$= \frac{1-\cos x}{\sin x},$$
$$= \frac{\sin x}{1+\cos x},$$
$$\cot\tfrac{x}{2} = \sqrt{\frac{1+\cos x}{1-\cos x}},$$

Writing Math

Consider the following three different ways to say the same thing,

1. Let C be a set of cities. The population size of a city $c \in C$ is noted $size(c)$. The average size of cities in C is,

$$m = \frac{\sum_{c \in C} size(c)}{|C|}$$

2. Suppose we have a set of n cities each one identified with an index $1 \leq i \leq n$. The population of the i-th city is noted s_i. The average size of the cities is,

$$m = \frac{1}{n} \sum_{i=1}^{n} s_i$$

3. Let C be a set of cities where each city is a pair $c = (id, s)$ with id being its name and s being its population. The average size of the cities in C is,

$$m = \frac{\sum_{(id,s) \in C} s}{n}$$

where $n = |C|$.

Mathematical Symbols

Mathematical expressions are made out of *numbers*, and *symbols*. Some symbols refer to *operators* (e.g. $+, -, \ldots$). Other symbols are used to name numbers, functions, sets,... These **symbols** are a letter in a given style. The usual symbol styles are:

- lower case italics: x, y, p, q
- lower case boldface: $\mathbf{x}, \mathbf{y}, \mathbf{p}, \mathbf{q}$
- upper case italics: P, Q, R
- upper case boldface: $\mathbf{P}, \mathbf{Q}, \mathbf{R}$
- Calligraphy: $\mathcal{P}, \mathcal{Q}, \mathcal{R}, \mathcal{X}$
- Greek alphabet (lower and upper case) $\alpha, \beta, \Delta, \Gamma$

Plain text

Warning

Plain text is never a valid style in mathematics because it is important to distinguish between the English and the Math. (e.g. It is wrong to write that "n is the number of elements in set C"; it is right to write that "n is the number of elements in set C").

There is one exception. Some standard functions (e.g. $y = \sin x, z = \max\{x, y\}, z = x \mod y$) are sometimes written in plain text precisely to emphasize them inside of a Math expression

Styles

Warning

In English, you can write a word in plain text, *italics* or **boldface**, lower or UPPER case and it is still the same word.
In math writing, if you define a symbol (e.g. n) the definition does not extend to other fonts or styles (e.g. $\mathrm{n}, \mathbf{n}, N, \mathcal{N} \ldots$).

Example:

- I left the keys on the table, but the TABLE was broken and the **keys** fell off
- In a function $f(x) = ax + b$, the value $\mathbf{a}$ is called the slope.

Styles

Hint

Do not to use the same letter in different, but similar, styles to denote different things, because it may be confusing.
For example, it is a bad idea to denote the size of a set of numbers m and their mean m.

The previous rule does not apply between lower and upper case. For instance, it is not unusual to name a set M and an arbitrary element m.

I would avoid naming a set C and an arbitrary element c because they look much more similar.

Notational Entity

Notational Entity

Upper case has greater *notational entity* than lower case, and calligraphic upper case has greater notational entity than upper case. Additional **notational entity** can also be obtained using boldface.

A notation of greater entity is to be used to name more complex elements.

Example:

- Let P be a set of integers
- Let p be an integer
- Let $\mathcal{P}$ be a partition of a set of integers

Notational Coherence

Notational Coherence

Use the same style to denote the same type of elements

Example:

- It would look strange that in the same document you use lower case and upper case italic to denote integers (e.g. let A_i and p_i denote the *area* and *population* of city i)
- ... unless there is a very important integer whose "name" you want to emphasize (e.g. let p_i denote the population of city i. Let P denote the sum of populations of all cities being considered)

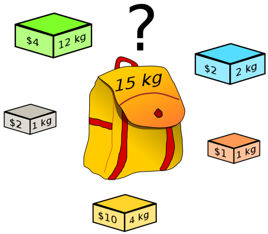
Modifiers

Modifiers

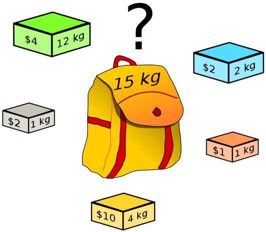
Sometimes we need to name several mathematical objects that are very related. To emphasize their relation one can use the same letter with a **modifier**. Modifiers are **sub-indexes**, **super-indexes**, **accents** and **primes**. These elements can be very useful in some situations, but one should use with care.

- **Accents and priming:** useful to denote variations (e.g. derivative $f(x), f'(x)$) or special elements (e.g. optimum x^*)
- **Indexes** are useful for enumeration (e.g. $X = \{x_1, x_2, \ldots, x_i, \ldots, x_n\}$) or identification ($s_i$ and w_i are the size and wealth of city i)

Example: Knapsack Problem

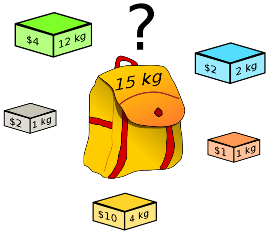


Example: Knapsack Problem



Consider a knapsack with capaticity K and set of n objects where p_i and w_i denotes the profit and the weight of object i. The problem is to chose a subset of objects fitting in the knapsack and maximizing their profit.

Example: Knapsack Problem



Consider a knapsack with capaticity K and set of n objects where p_i and w_i denotes the profit and the weight of object i. The problem is to chose a subset of objects fitting in the knapsack and maximizing their profit.

The last sentence can be rephrased as,

- maximize $\sum_{i=1}^{n} p_i x_i$
- subject to $\sum_{i=1}^{n} w_i x_i \leq K$
- with $x_i \in \{0, 1\}$

Words as symbols

Warning

For complex definitions (specially functions) words are sometimes a good idea, because they are easy to memorize.

- E.g. sin for *sinus*, max for *maximum*
- E.g. $MC(G)$ set of maximal cliques of a graph G

Computer Scientists use words too often (probably because of our habit to name variables and algorithms). So it is good advise to restrict our notation to single letter symbols unless there is a good reason against it.

Notational Conventions

Some usual naming conventions:

- **scalars** a, b, x, y, p, q
- **indexes** i, j, k
- **vectors** (i.e, fixed-size ordered sequences) $\vec{v} = (v_1, v_2, \ldots, v_n)$.
- In Computer Science we often talk about **arrays** and **lists**, which are finite (but not necessarily fixed-size) ordered sequences. Usual notation for arrays is $s = (s[1], s[2], \ldots, s[n])$, and for lists is $l = (l_1, l_2, \ldots, l_n)$
- **sets**, **multisets** $S = \{x_1, x_2, \ldots, x_n\}$
- **tuples** (fixed-size sequences of heterogeneous elements) $P = (A, B)$
- **functions** f, g, h
  - $f : A \longrightarrow B$ (that means $x \in A$, then $f(x) \in B$)
  - $f : A \times A \longrightarrow B$ (that means $x, y \in A$, then $f(x, y) \in B$)

Some hints from Donald Knuth

- Facilitate memorization (e.g. $\vec{v}$ for vectors, p for points, S for sets, ...)
- Avoid **unnecessary subscripts** (e.g. x and y is better than x_1 and x_2, or x and x')
- Avoid **piling subscripts** (e.g. $x_{i'}^{p_j}$)
- Avoid **unfamiliar symbols** (e.g. Ξ, Θ) and **unfamiliar accents** (e.g. $\hat{a}$, $\check{a}$, $\dot{a}$)

Some Advanced Operators

Summation

$$\sum_{it} expr_{it}$$

Example:

- $\sum_{i=1}^{n} a_i$
- $\sum_{a \in S} a$
- $\sum_{i=1}^{n} \sum_{j=1}^{m} (a_i + b_j)$

Similar syntax for: $\prod$, max, min, $\forall$, $\exists$, etc

Some Advanced Operators

Set constructor

Allows to define a set based on a previous set and filtered with a condition

$$\{domain|\ filter\ condition\}$$

Example:

- $\{a \in \mathbb{N}|\ \exists_{0<b<a}\ \ b^2 = a\}$
- $\{(a, b) \in A \times B|\ a < b\}$

On "efficient" definitions

What is better?

- $\{a \in \mathbb{N} | \ \exists_{0<b<a} \ \ b^2 = a\}$ because it is simpler
- $\{a \in \mathbb{N} | \ \exists_{0<b\leq a/2} \ \ b^2 = a\}$ because it is more efficient

On "efficient" definitions

What is better?

- $\{a \in \mathbb{N} | \ \exists_{0<b<a} \ \ b^2 = a\}$ because it is simpler
- $\{a \in \mathbb{N} | \ \exists_{0<b\leq a/2} \ \ b^2 = a\}$ because it is more efficient

.

This is not an algorithm, you silly. When defining something you look for simplicity to make the definition as easy to grasp as possible. The very concept of efficiency does not even make sense in this context.

I like better this one:

- $\{a \in \mathbb{N} | \ \exists_{b\in\mathbb{N}} \ \ b^2 = a\}$

Case-based definitions

Case-based definitions

Allows to define based on parameter conditions

$$parameterizedname = \begin{cases} expr1, & cond1 \\ expr2, & cond2 \\ ... & \\ exprn, & condn \end{cases}$$

Conditions must be mutually exclusive and their union must cover all posible cases (this is why very often the last condition is "otherwise")

Example:

$$f(x) = \begin{cases} x^2, & -100 \leq x \leq 100 \\ |x|, & \text{otherwise} \end{cases}$$

Example

Suppose we want to define **absolute value** $|x|$. Which definition do you prefer?

- $|x| = +\sqrt{x^2}$
- $|x| = \begin{cases} x, & x \geq 0 \\ -x, & \text{otherwise} \end{cases}$

Examples

The following excerpts are from real papers that I have reviewed. They contain many aspects that can be improved, ranging from obvious mistakes to style details.
So your goal is to identify these improvable aspects.

Example

A graph $G = (V, E)$ consists of a vertex set $V = \{v_1, v_2, \ldots, v_n\}$ and an edge set $E \subseteq \{(u, v) \mid u, v \in V\}$, where the vertices represent variables and the edges represent the dependencies between variables. If two distinct vertices u and v are connected by an edge, they are said to be adjacent in G, denoted as $u \sim v$. For any vertex $\alpha \in V$, its neighborhood set is defined as $ne_G(\alpha) = \{v \in V \mid v \neq \alpha, v \sim \alpha\}$; for any subset $A \subseteq V$, its neighborhood set is $ne_G(A) = \left(\bigcup_{\alpha \in A} ne_G(\alpha)\right) \setminus A$. To simplify the notation, we use $ne(\alpha)$ and $ne(A)$. Specifically, let $N[v] = ne(v) \cup \{v\}$ and $N[A] = ne(A) \cup A$. If all edges in G are undirected, it is called an undirected graph; if all edges are directed, it is called a directed graph; otherwise, it is called a mixed graph.

Things that are not OK

- different notation for vertices (v_i, v, u, α)
- In sentence starting *If two distinct ...*, connected is not defined. Rephrase as *If there is an edge between two vertices u and v, we say that they are adjacent, and this is noted as $u \sim v$*
- Sentence starting *Specifically...*, needs rephrasing. Also Notation $N[]$ is not consistent with $ne()$
- Sentence starting *If all edges..* is confusing (what does it mean to be undirected?)
- there are concept that are defined and not used later (e.g. adjacent)
- Through the text, each time a concept is defined, emphasize (in italics or boldface) the name of what is being defined (e.g. **adjacent**, **neighborhood**,...)

Example cont.

A node labeling α of the graph $G = (V, E)$ is a bijection from $V(G)$ to $\{1, 2, \ldots, n\}$, where $\alpha(v)$ denotes the label of vertex v, and the node labeling is often represented as $\alpha = (v_1, \ldots, v_n)$, with $\alpha(v_i) = i$ for $1 \leq i \leq n$. Given a node labeling α, the monotonic neighborhood of node x is defined as $madj(x) = \{y \in V \mid y \in ne(x), \alpha(y) > \alpha(x)\}$, and we denote $\mathcal{L}_i = \{v_i, v_{i+1}, \ldots, v_n\}$. Specifically, in a Directed Acyclic Graph (DAG), if $x \to y$, then $\alpha(x) < \alpha(y)$.

Things that are not OK

- α was already used
- What is $V(G)$? (I know that they mean *the set of vertices of graph* G, but the reader does not have to guess)
- The definition of $\alpha = (...)$ is not clear
- Definition of monotonic neigh. not clear, notation $madj()$ is ugly.
- Definition of $\mathcal{L}$ not clear and why calligraphy upper case style?
- $x \rightarrow y$ is not defined

Example cont.

A subgraph $G(V') = (V', E')$ of $G = (V, E)$ is induced by a subset $V' \subseteq V$, where $E' = \{(X, Y) \in E \mid X \in V', Y \in V'\}$. If every pair of vertices in an undirected graph G is connected by an edge, then G is called a complete graph. If $G(K)$ is a maximal complete subgraph of G (i.e., $G(K)$ is complete, and for any W such that $K \subseteq W \subseteq V$, $G(W)$ is not complete), then $G(K)$ is called a clique of G.

Things that are not OK

- Now vertices are named in upper case (X, Y)?
- Definition of complete graph and clique is not the standard one.

Example

2 Definitions and Preliminaries

Note that all the lemmas presented in this section can be found in a book on treewidth by Ton Kloks [Kloks, 1994]. All the graphs used in this paper are finite, undirected, connected and simple. We denote an undirected graph by $G(V_G, E_G)$ where V_G is the set of vertices of the graph and E_G is the set of edges of the graph. The *neighborhood* of a vertex v denoted $N_G(v)$ is the set of vertices that are neighbors of v. A chord of a cycle C is an edge not in C whose endpoints lie in C. A *chordless cycle* in G is a cycle of length at least 4 in G that has no chord.

Definition 2.1. *A graph is* **chordal** *or triangulated if it has no chordless cycle.*

A graph $G' = (V', E')$ is called a *subgraph* of G if $V' \subseteq V$ and $E' \subseteq E$. If $W \subseteq V$ is a subset of vertices then $G[W]$ denotes the subgraph induced by W. A clique is a graph G that is completely connected. A *triangulation* of a graph G is a graph H such that G is a subgraph of H, H has the same set of vertices as G and H is chordal.

Things that are not OK

- Defines Neighborhood assuming neighbor is known.
- Definition of *chordless* is obvious and silly ("colorless" (without color), "homeless" (without a home), "useless" (without use or value),...)
- Definition of *chordless* is a double negation. That this is a bad idea becomes clear in Definition 2.1, where *no chordless* again is a double negation.
- As a matter of fact, if you think a little about it, chordal means that every cycle of size ≥ 4 has a chord
- Definition of $G[W]$ is wrong because it does not indicate what the edges are
- The last paragraph does not flow. It is messy.

Examples

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In the following examples, we are going to see a concept informally and our task is to write a formal definition of the concept. It may require to decide the right notation.
Many of the notions that we are going to consider are invented and completely useless in practice. We only introduce them for practicing purposes

Consider a lists $l = (l_1, l_2, \ldots, l_n)$.

1. the **reverse** of l is ,,,
2. the 1-shift of l is ,,,
3. **capping** a list l with k is ,,,

Examples

Definition (Reverse)

The **reverse** of list $l = (l_1, l_2, \ldots, l_n)$ is a list $r = (r_1, r_2, \ldots, r_n)$ such that $r_i = l_{n-i+1}$

Definition (1-shift)

The 1-**shift** of list $l = (l_1, l_2, \ldots, l_n)$ is a list $r = (r_1, r_2, \ldots, r_n)$ such that

$$r_i = \begin{cases} l_{i-1}, & i > 1 \\ l_n, & i = 1 \end{cases}$$

Definition (Capping)

Capping list $l = (l_1, l_2, \ldots, l_n)$ with k is a list $r = (r_1, r_2, \ldots, r_n)$ such that $r_i = \min\{l_i, k\}$

Examples

Consider a lists $l = (l_1, l_2, \ldots, l_n)$.

1. the peak of l is ...
2. the peak-location is...
3. the symmetry density of l ...
4. the repetition number of l is ...

Examples

Definition (peak)

The **peak** of a list $l = (l_1, l_2, \ldots, l_n)$ is the maximum among its elements. Formally, $\max_{1 \leq i \leq n}\{l_i\}$

Definition (peak-place)

The **peak-place** of a list $l = (l_1, l_2, \ldots, l_n)$ is the position of its peak. Formally, $\mathrm{argmax}_{1 \leq i \leq n}\{l_i\}$

Definition (Symmetry density)

The **symmetry density** of a list $l = (l_1, l_2, \ldots, l_n)$ is, $\frac{|\{i \in 1..(n+1)/2 |\ l_i = l_{n-i+1}\}|}{(n+1)/2}$

Definition (Repetition Factor)

The **repetition factor** in $l = (l_1, l_2, \ldots, l_n)$ is $\frac{|\{l_i \in l \mid \exists_{j \neq i}\ l_i = l_j\ \}|}{n}$

Examples with sets

Given two sets of integers A and B,

1. the elements of A whose square is in B is...
2. the pairs whose first and second element is in A and B, respectively that are no further apart than k is...
3. the set of even numbers common to both sets is...
4. the set of subsets of A whose sum is 0 is...
5. the set of subsets of A that are also subsets of B is...

Examples with sets

Given two sets of integers A and B,

1. the elements of A whose square is in B is, $\{a \in A \mid a^2 \in B\}$
2. the pairs whose first and second element is in A and B, respectively that are no further apart than k is, $\{(a, b) \in A \times B \mid |a - b| \leq k\}$
3. the set of even numbers common to both sets is, $\{a \in A \cap B \mid (a \bmod 2) = 0\}$
4. the set of subsets of A whose sum is 0 is, $\{P \subseteq A \mid \sum_{p \in P} p = 0\}$
5. the set of subsets of A that are also subsets of B is, $\{P \mid P \subseteq A \cap B\}$

Examples with matrices

Consider a matrix A of dimension $n \times n$ where a_{ij} denotes the element in row i and column j

1. the vector containing the maximum of each row is ...
2. the minimum among the maximum of each row is ...
3. the elements in A that are equal to their symmetric (the symmetry is defined with respect to the matrix diagonal) is ...
4. the elements that are larger than any of its neighbors is...

Examples with matrices

Consider a matrix A of dimension $n \times n$ where a_{ij} denotes the element in row i and column j

1. the vector containing the maximum of each row is $\vec{v} = (v_1, v_2, \ldots, v_n)$ with $v_i = \max_{1 \leq j \leq n} a_{ij}$
2. the minimum among the maximum of each row is $\min_{1 \leq i \leq n}\{\max_{1 \leq j \leq n} a_{ij}\}$

Examples with matrices

Consider a matrix A of dimension $n \times n$ where a_{ij} denotes the element in row i and column j

1. the elements in A that are equal to their symmetric (the symmetry is defined with respect to the matrix diagonal) is $\{(i,j), 1 \leq j < i \leq n |\ a_{ij} = a_{ji}\}$
2. the elements that are larger than any of its neighbors are,

$$\{(i,j), 1 \leq i, j \leq n |\ a_{ij} > b \text{ for all } b \in N(a_{ij})\}$$

   where,

$$N(a_{ij}) = \{a_{i-1,j}, a_{i+1,j}, a_{i,j-1}, a_{i,j+1}\}$$

   where we assume for convenience $a_{0j} = a_{i0} = 0$

Examples with graphs

Consider an undirected graph $G = (V, E)$

1. A path is ...
2. A loop is ...
3. The graph's shortest loop is ...
4. The graph embedded by $U \subset V$ is ...
5. Graph $G = (V, E)$ is a clique if ...
6. The MaxClique problem is...
7. Graph $H = (W, A)$ is an isomorphism of $G = (V, E)$ if ...

Examples with graphs

Consider an undirected graph $G = (V, E)$

1. A path is a sequence of vertices $(u_1, u_2, \ldots, u_k)$ such that $(u_i, u_{i+1}) \in E$ for all $1 \leq i < k$
2. A loop is a path $(u_1, u_2, \ldots, u_k)$ such that $k > 1$ and $u_1 = u_k$
3. The graph's shortest loop is (just define the lenght)
4. The graph embedded by $U \subset V$ is $G' = (U, E')$ with $E' = \{(u, w) \in E |\ u \in U, w \in U\}$
5. Graph $G = (V, E)$ is a clique if $E = V \times V$
6. Given a graph $G = (V, E)$ the MaxClique problem consists on finding the largest (with respect to the number of vertices) embedded clique
7. Graph $H = (W, A)$ is an isomorphism of $G = (V, E)$ if there is a mapping (i.e, bijection) $f : V \to W$ such that $(u, w) \in E$ if and only if $(f(u), f(w)) \in A$